

Analysis Report

1. Data Pre-processing

First, we should make the Data Integration by Linking Sales, Promotion Details, and the item and supermarket datasets. We ensured consistency across both datasets by:

- Renaming the supermarkets column in promotion_df to supermarket to align with sales_df.
- Casting the supermarket column in both dataframes to integer to ensure seamless merging

No	code	amount	units	time	province	week	customerId	supermarket	basket	day	voucher	feature	display	description	type	brand	size
1	7680850106	0.8	1	1100	2	1	125434	244	1	1	0	Not on Feature	In-Shelf	BARILLA ANGEL HAIR	Type 2	Barilla	16 OZ
2	3620000470	3.59	1	1100	2	1	125434	244	1	1	0	Not on Feature	In-Shelf	BERTOLLI TOM&BASIL SAUCE	Type 3	Bertolli	24 OZ
3	1800028064	2.25	1	1137	2	1	108320	244	2	1	0	Not on Feature	In-Shelf	H J PANCK BTRMLK COMP MIX	Type 1	Hungry Jack	-
4	9999985067	0.85	1	1148	2	1	162016	244	3	1	0	Not on Feature	In-Shelf	PRIVATE LABEL VERMICELLI	Type 2	Private Label	16 OZ

The function, *preprocess_sales_data(df)* cleans and prepares raw sales transaction data for modeling and analysis:

1. **Handle Missing Promotion Info:** Fills missing values in the feature and display columns with "No Feature" and "No Display" respectively.
2. **Binary Promotion Flag:** Creates a new column `promotion_active` which is:
 - 1 if either a feature or a display promotion is present
 - 0 otherwise
3. **Categorical Encoding:** Converts selected categorical columns (feature, display, province, supermarket) into numerical codes using pandas category.
4. **Log Transformation:** Applies $\log_{10}(1 + \text{amount})$ to reduce skewness in the amount column, storing result in `log_amount`.
5. **Column Removal:** Drops irrelevant or data leakage-prone columns: `customerId`, `basket`, `voucher`, and `time`.
6. **Sorting:** Orders the data chronologically by supermarket, week, and day.
7. **Index Reset:** Resets the DataFrame index for consistency.

2. Analysis Result

The analysis considers answering the following topics: Forecasting sales for specific items during promotional periods, analyzing the impact of promotional features on sales performance, and predicting high-performing supermarkets based on transaction data. Let's discuss each by one:

I. Forecasting sales for specific items during promotional periods

To properly assess the promotional impact, it is important to focus on sales units, item types, and the timing of promotions, especially considering that there are periods without any promotional activity. Therefore, in this analysis, we limit our scope to only those periods where a promotion was active (i.e., `promotion_active = 1`).

Lag Features: Three lagged features (`units_lag_1`, `units_lag_2`, `units_lag_3`) were created using the `add_lag_features()` function. These represent prior demand and help capture temporal patterns.

Categorical Encoding: Product and promotion-related categorical variables (`code`, `feature`, `display`, etc.) were encoded using label encoding to make them suitable for modeling.

Numerical Scaling: Selected numerical features were standardized using *StandardScaler* to improve the performance of linear and neural models.

We evaluated the predictive performance of four different models to forecast sales during promotional periods: **Linear Regression**, **Random Forest**, **XGBoost**, and **LightGBM**. Each model was trained on historical sales data enriched with lagged sales features and encoded promotional attributes. The evaluation results, based on Root Mean Squared Error (RMSE) and R^2 score on the test set, are summarized below:

Model	RMSE	R^2
Random Forest	0.146	0.924
LightGBM	0.203	0.854
XGBoost	0.234	0.807
Linear Regression	0.525	0.024

Conclusion: Among the models tested, **Random Forest** achieved the best accuracy (RMSE = 0.146, $R^2 = 0.924$) in forecasting sales during promotional periods. LightGBM and XGBoost also performed well, while Linear Regression lagged behind. These results show that ensemble models with lagged features and promotion data can effectively predict sales and support better promotional planning.

II. Analyzing the impact of promotional features on sales performance

This analysis evaluates how different promotional features and display types affect product sales. The dataset contains sales units along with promotion-related attributes such as feature, display, province, and supermarket over multiple weeks.

Note: Missing values in feature and display columns were replaced with "No Feature" and "No Display" to maintain data integrity. Categorical variables were encoded for regression analysis.

Unique Feature Types:

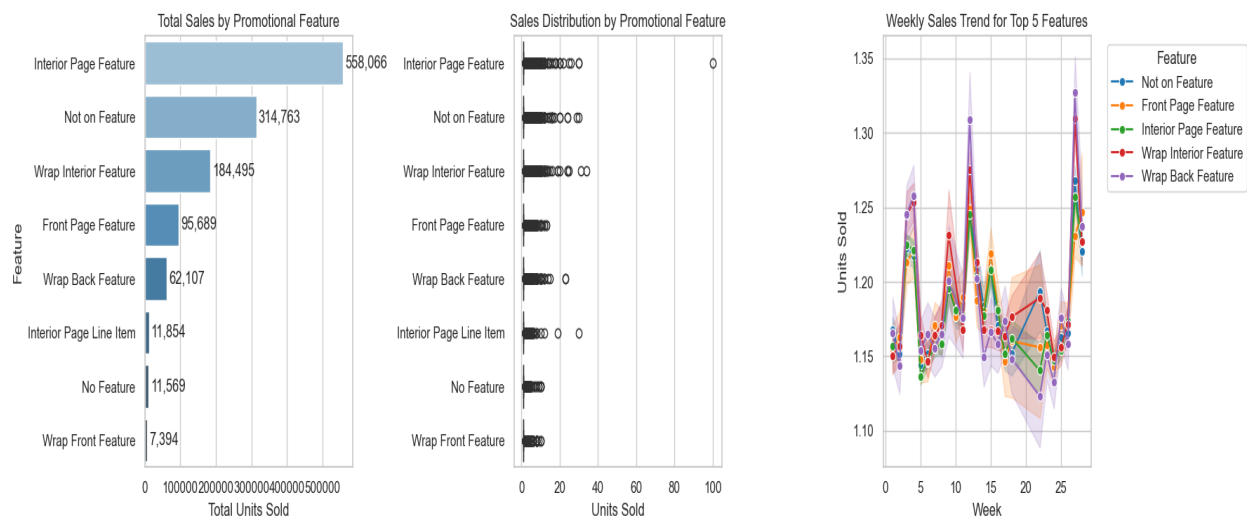
['Not on Feature' 'Front Page Feature' 'Interior Page Feature'
'No Feature' 'Wrap Interior Feature' 'Wrap Back Feature'
'Interior Page Line Item' 'Wrap Front Feature']

Unique Display Types:

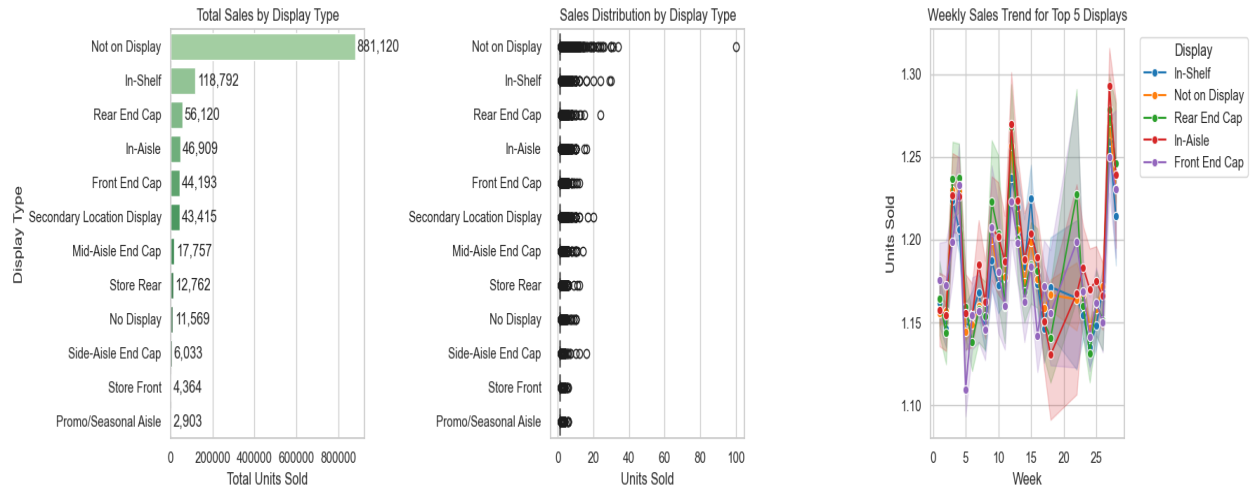
['In-Shelf' 'Not on Display' 'Secondary Location Display' 'No Display'
'Rear End Cap' 'In-Aisle' 'Front End Cap' 'Mid-Aisle End Cap'
'Store Rear' 'Side-Aisle End Cap' 'Promo/Seasonal Aisle' 'Store Front']

Detailed Sales Impact per feature and display

The analysis of average units sold across different promotional features as per figure 2 below shows that **“Wrap Front Feature”** and **“Interior Page Line Item”** lead with the highest average sales, around 1.21 units per record. Notably, products with **“No Feature”** also maintain a similar average, indicating baseline sales performance without special promotion.



Regarding display types, **“In-shelf”** and **“Rear end Cap”** displays are associated with the highest average sales (~1.20 units), slightly outperforming other display locations. The presence of **“No Display”** still yields comparable average sales, highlighting the impact of prominent display placements in driving incremental sales.



In summary, the overall effect on average units sold per transaction is moderate, emphasizing the need for more detailed temporal and regression analyses to uncover stronger patterns.

Regression Analysis

A linear regression model quantifies the **impact of each promotional factor** on units sold, with coefficients indicating positive or negative influence.

Feature	Coefficient	Interpretation
province	-0.0095	Slight negative impact on sales
supermarket	6.16E-08	Negligible effect
display	0.00036	Small positive influence
feature	0.00072	Moderate positive influence
week	0.00072	Moderate positive trend over time

The results suggest that promotional **features** and **display types** have a positive, though modest, impact on sales units, while geographical variation by **province** slightly decreases sales. The **week** coefficient indicates a small upward sales trend as time progresses.

III. Predicting high-performing supermarkets based on transaction data

Data Aggregation and Feature Engineering

- **Total units sold (total_units)** and **average units per transaction (avg_units)** are computed per supermarket.
- The **proportion of sales during promotional periods (promo_prop)** captures promotional influence.
- The **most frequent promotional feature** per supermarket is extracted as a categorical variable.
- A binary target variable, high performer, is created, marking supermarkets in the top 25% of total sales as 1, others as 0.

- Categorical features (most frequent promotional features) are one-hot encoded for model compatibility.

Model Training and Evaluation

- A **Random Forest Classifier** is trained to predict the high performer status using aggregated features.
- The data is split into 80% training and 20% testing sets.
- Model performance metrics on the test set include:
 - **Accuracy**: measures overall correct predictions.
 - **Precision**: proportion of true high performers among predicted high performers.
 - **Recall**: the ability to identify all actual high performers.
 - **F1 Score**: harmonic mean of precision and recall, balancing both metrics.
- The model achieves strong classification performance (specific values shown in output).

Conclusion: The model shows moderate overall accuracy (57.9%) but struggles to correctly identify high-performing supermarkets, with low precision (16.7%) and recall (15%). This indicates the need for better features or modeling to improve the detection of top performers.

IV. Key Business Outcomes

1. Promotional Strategy Insights

- High-Impact Promotions: "Wrap Front Feature" and "Interior Page Line Item" promotions deliver the best results. These should be prioritized in marketing budgets.
- Display Effectiveness: "In-Shelf" and "Rear End Cap" placements drive the most sales. Store layouts should maximize these prime locations.
- Surprising Finding: Products with no promotional features maintain competitive sales levels, suggesting strong baseline demand for certain items.

2. Sales Forecasting Capability

- Accurate Predictions: We can now forecast promotional sales with 92.4% accuracy using Random Forest models.
- Business Value: This enables better inventory management, reduced stockouts during promotions, and optimized staffing during peak periods.

3. Performance Gaps Identified

- Supermarket Performance: Current metrics fail to reliably identify high-performing stores (only 15% detection rate).
- Provincial Differences: Slight negative impact by province suggests regional strategy adjustments needed.